CS383 · SOFTWARE ENGINEERING

Midterm 472 Practice

20 marks

Muharib

01 Questions

Q1 MCQ · 1 – 4 · 8 M

Multiple-choice questions (each question = 0.5 mark)

1. Which of the following is one of the main duties of a software project manager?
   A. Writing source code for every module personally
   B. Estimating costs, scheduling tasks, and monitoring project development
   C. Performing all unit testing on behalf of the team
   D. Making all technical design decisions without team input
2. A successful software project should:
   A. Deliver all planned features regardless of the deadline or cost
   B. Be delivered on time, within budget, and meet the agreed requirements
   C. Minimize cost above every other criterion
   D. Maximize the number of customer meetings during development
3. Which item is normally included in a project plan?
   A. Sample source code for each task
   B. Risk analysis, resource requirements, and a work breakdown
   C. A list of developer salaries
   D. The product's marketing strategy
4. Why should project tasks be self-contained where possible?
   A. They are harder to test in isolation
   B. They increase coupling between tasks
   C. They are easier to estimate, manage, and test independently
   D. They force a fixed order of development

Q1 MCQ · 5 – 8

5. Which of the following best qualifies as a project milestone?
   A. Fixing a single syntax error in a module
   B. Completion and formal sign-off of the requirements specification
   C. Installing a web browser on the developer's machine
   D. Adding a comment to the project wiki
6. What is the main purpose of risk management in software project management?
   A. To replace the need for detailed planning
   B. To identify risks, assess their impact, and plan responses before problems occur
   C. To increase the project budget automatically
   D. To substitute for the testing phase
7. Which of the following is part of people management?
   A. Selecting team members and assigning suitable roles
   B. Removing stakeholder requirements to save effort
   C. Replacing the project schedule with source code
   D. Cancelling performance reviews to build trust
8. A team is developing a safety-critical avionics control system with strict regulatory certification and well-defined, rarely changing requirements. Which process model is most appropriate?
   A. Scrum with two-week sprints and no formal documentation
   B. A plan-driven (waterfall / V-model) process with full documentation and formal reviews at each stage
   C. Ad-hoc coding without a defined process
   D. Big-bang integration at the very end of development

Q1 MCQ · 9 – 12

9. Why should task dependencies be minimized in project planning?

A. To reduce the need for project monitoring

B. To enable parallel work and limit the impact of task delays

C. To eliminate the need for documentation

D. To reduce communication between team members

10. A project manager is preparing the plan for a new university registration system. Which set of activities is the most appropriate for project planning?

A. Writing source code, debugging modules, and deploying the final system

B. Setting milestones, breaking work into tasks, estimating time and resources, and minimizing task dependencies

C. Drawing only UML diagrams and ignoring scheduling constraints

D. Focusing only on team motivation without defining deliverables

11. A functional requirement is best described as:

A. A constraint on project cost only

B. A service or behavior that the system must provide

C. The choice of programming language used in the project

D. A project milestone or deadline

12. Which of the following is a non-functional requirement?

A. The system shall allow students to reset their passwords

B. The system shall generate a transcript as a PDF document

C. The system shall respond to any search request in less than 2 seconds

D. The system shall allow advisors to approve registration requests

Q1 MCQ · 13 – 16

13. User requirements are usually:
    A. Expressed only as source code
    B. High-level statements understandable by non-technical stakeholders
    C. More detailed than system requirements
    D. Written only by developers for developers
14. Which statement is the best example of an ambiguous requirement?
    A. The system shall store each student's ID, name, and email
    B. The system shall be user-friendly and fast
    C. The system shall lock an account after three failed login attempts
    D. The system shall export reports in PDF format
15. In requirements engineering, validation mainly checks whether the requirements are:
    A. Syntactically well-formatted in a single language
    B. Valid, consistent, complete, realistic, and verifiable
    C. Implemented in an object-oriented language
    D. Partitioned into independent modules
16. Which process model is generally most suitable when requirements change frequently and customer feedback is continuous?
    A. Waterfall
    B. Agile
    C. Pure plan-driven
    D. Big-bang integration

Q2 USER / SYSTEM · FR / NFR · 8 M

For each statement below, classify (i) User vs. System requirement and (ii) Functional (FR) vs. Non-functional (NFR). (each row = 2 marks)
Reminder: User requirements = high-level, stakeholder-facing statements. System requirements = detailed technical statements describing what the system will do or constrain.

Requirement	User / System	FR / NFR
The system should be easy for new students to learn within one lab session.		
The system shall allow an instructor to upload course files in PDF format.		
The application shall support at least 500 concurrent users during registration.		
The system shall encrypt all stored passwords using the SHA-256 algorithm.		

Q3 SHORT ANSWER & CALCULATION · 4 M

A. Process model selection (1 mark)

A small startup team of five developers is building a mobile application. The customer gives weekly feedback, features are re-prioritized frequently, and an early working version is required quickly.

Tick one: ☐ Agile ☐ Plan-Driven ☐ Component-Based

One reason (max. one sentence):

B. Agile planning – sprint velocity (1 mark)

A team completed 18, 22, 20, and 24 story points in the last four sprints. Compute the average sprint velocity.

C. Task scheduling (2 marks)

A software project has four tasks with estimated durations of 5, 8, 4, and 7 days.

(i) Compute the total estimated duration if all tasks are performed sequentially.

(ii) If Task 3 can be executed in parallel with Task 2 (and Tasks 1 and 4 remain sequential), what is the new total duration?